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Continuous-Frequency Characterization of the Temperature Coefficient of Phase Shift of Photoreceiver in Spaceborne Gravitational Wave Detection

Author Information**Author names:**

Zekai Huang, Baifu Lu, Yurong Liang*

Affiliations:

National Gravitation Laboratory, MOE Key Laboratory of Fundamental Physical Quantities Measurement, and School of Physics, Huazhong University of Science and Technology, Wuhan 430074, China

d202580039@hust.edu.cn (Zekai Huang); ltt1999oct@gmail.com (Baifu Lu); liangyurong20@hust.edu.cn (Yurong Liang)

Corresponding author:

Yurong Liang

E-mail: liangyurong20@hust.edu.cn

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Abstract: Phase stability of photoreceiver front-end circuits is important in precision heterodyne interferometry, including space-borne gravitational wave detection and intersatellite laser ranging. Conventional characterization schemes only measure the temperature coefficient of phase shift (TCPS) at a single discrete frequency, whereas practical photodetectors work within a wide bandwidth such as from 5 MHz to 25 MHz. How to accurately acquire the frequency-dependent TCPS over the full operating bandwidth remains an open technical challenge. In this work, a frequency-continuous TCPS measurement method combining temperature modulation and frequency sweeping is proposed. The phase-frequency responses of the device under test are measured at different steady-state temperatures using a vector network analyzer for frequency sweeping, and the TCPS is obtained by phase differencing between adjacent temperature points. The method is experimentally validated over 1–50 MHz using a bipolar-transistor-based transimpedance amplifier as a representative photoreceiver front-end. The measured TCPS exhibits clear frequency dependence and reaches an extremum of -7.4 ± 0.2 mrad/K at 19.8 ± 0.2 MHz. Monte Carlo simulations considering circuit-parameter uncertainties show consistency with the measured extremum frequency and magnitude. The proposed method provides a useful tool for evaluating thermally induced phase drift in photoreceiver front-ends for precision laser interferometry and other related areas.

Keywords

Photoreceiver front-end; Transimpedance amplifier; Heterodyne interferometry; Temperature coefficient of phase shift; Frequency-domain measurement

1. Introduction

High-precision laser interferometry and optoelectronic phase readout techniques are widely used in space-based gravitational wave detection, inter-satellite laser ranging, precision displacement measurement, and optical sensing [1-7]. Specifically, for space-based gravitational wave detection, inter-satellite laser interferometers detect gravitational waves in the low-frequency band of 0.1 mHz~1 Hz. In such systems, the physical quantity to be measured is typically converted into phase variation of heterodyne beat-note signals. The long-term stability of phase readout directly determines the measurement sensitivity of the system [2]. For example, in space-based gravitational wave detection, inter-satellite laser interferometry requires highly stable phase measurements of MHz-level heterodyne beat-note signals, and consequently, phase noise in the optoelectronic readout link directly affects the system performance [8, 9]. Therefore, the phase stability of the optoelectronic readout chain is a critical issue in the design and performance evaluation of precision heterodyne interferometric systems.

In a laser heterodyne interference measurement link, the beat-note optical signal generated by interference must be converted into an electrical signal by a photoreceiver before it can be processed by the subsequent phase readout system [8]. As the front-end circuit of the photoreceiver, the transimpedance amplifier (TIA) converts the weak photocurrent generated by the photodiode into a usable voltage signal. Its transimpedance gain, bandwidth, phase-frequency response, and noise characteristics affect both the amplitude stability and phase stability of the beat-note signal [10, 11]. In practice, temperature fluctuations can cause drifts in the feedback resistance and capacitance, the small-signal parameters of active devices, and parasitic parameters, thereby changing the phase-frequency response of the TIA and affecting the final output phase [12, 13]. To quantitatively describe the sensitivity of the circuit phase to temperature variations, the temperature coefficient of phase shift (TCPS) is commonly introduced, which is defined as the output phase variation induced by a unit temperature change. Within a small temperature variation range, thermally induced phase noise can be estimated from the product of the ambient temperature noise and the TCPS.

Previous studies have investigated temperature-related phase drift in photoreceivers, TIAs, and phase readout systems. In 2014, Gerberding equivalently simulated temperature-induced circuit-parameter variations by changing the feedback resistance of a TIA, and theoretically analyzed the resulting change in the phase-frequency response. The results showed that, at an operating frequency of 20 MHz, an approximately $\pm 2\%$ variation in a 10 k Ω feedback resistance could induce an output phase shift of about 10 mrad [14]. In 2017, Fernández Barranco et al. experimentally investigated the photoreceivers used in the GRACE Follow-On inter-satellite laser interferometry system. By applying temperature perturbations to the photoreceiver and measuring the corresponding phase variation of the beat-note signal, they obtained the experimental TCPS. The measurements showed that, at an operating frequency of 16 MHz, the single-channel TCPS was approximately 4 mrad/K, whereas the differential TCPS was approximately 100 μ rad/K [15]. In 2024, Feng et al. used a phasemeter to apply local temperature modulation to temperature-sensitive devices and synchronously measured the temperature and phase variations, thereby obtaining the TCPS of several devices at a single operating frequency [16].

The above studies indicate that temperature-to-phase drift is a critical concern in high-precision optoelectronic readout links, and the TCPS is an important parameter for evaluating circuit thermal stability and low-frequency phase noise. However, existing analyses and experimental results mostly focus on specific operating frequencies, typically providing the TCPS at a single frequency. In heterodyne optoelectronic readout links, the beat-note signal frequency is generally not fixed but varies over a certain range. Therefore, the response characteristics over the entire operating bandwidth need to be considered. A TCPS obtained at only a single frequency cannot fully reflect the thermally induced phase response of the photoreceiver front-end over the target frequency range, nor can it identify the frequency regions and extremum features where the temperature-induced phase response is more sensitive.

Based on this, a method is proposed for measuring the TCPS using a vector network analyzer (VNA) for frequency-sweeping combined with temperature

modulation. As a high-precision testing device based on scanning measurement technology, a VNA can simultaneously obtain the phase information of the measured network through short scan, with the core advantage of fast and accurate frequency and phase measurement. The method leverages these capabilities of the VNA to measure the phase-frequency response of the DUT at different steady-state temperatures, and derives the TCPS curve over the target frequency band from the phase differences between adjacent temperature points. Compared with single-frequency measurement of the TCPS, the proposed method can directly obtain the frequency-dependent distribution of the coefficient and identify the sensitive frequency regions and extremum features of the thermally induced phase response. A bipolar-transistor-based TIA is employed as a representative photoreceiver front-end to experimentally validate the proposed method. The proposed approach can be used for thermal stability evaluation, phase-noise budget analysis, and circuit-parameter optimization in high-precision optoelectronic readout chains.

2. Measurement object and method

2.1. TCPS definition

The output phase of the TIA depends not only on the input signal frequency but also on the ambient temperature and the operating state of the circuit. As research shows, temperature fluctuations will affect the performance parameters of the TIA, which in turn impacts its output phase. Therefore, the circuit output phase can be expressed as a function jointly determined by frequency, ambient temperature and circuit operating state. temperature:

$$\varphi = \varphi(f, T) \quad (1)$$

where f denotes the frequency of the test signal and T denotes the circuit temperature. When the temperature changes, the circuit output phase also varies accordingly. If the temperature changes from T to $T + \Delta T$, the resulting phase variation can be expressed as:

$$\Delta\varphi(f) = \varphi(f, T + \Delta T) - \varphi(f, T) \quad (2)$$

To quantitatively describe the sensitivity of the circuit phase to temperature variations, the TCPS is defined as the rate of change of phase with respect to temperature:

$$S_T(f) = \frac{\partial\varphi(f, T)}{\partial T} \quad (3)$$

where $S_T(f)$ represents the phase variation induced by a unit temperature change at frequency f , with a unit of rad/K.

In practical measurements, the continuous partial derivative of phase with respect to temperature cannot be obtained directly. Therefore, the TCPS is approximated using the phase difference over a finite temperature interval. For two adjacent steady-state temperature points, T_i and T_{i+1} , the corresponding phase-frequency responses are $\varphi(f, T_i)$ and $\varphi(f, T_{i+1})$, respectively. The TCPS within this temperature interval can then be expressed as:

$$S_{T,i}(f) = \frac{\varphi(f, T_{i+1}) - \varphi(f, T_i)}{T_{i+1} - T_i} \quad (4)$$

where $S_{T,i}(f)$ denotes the TCPS over the temperature interval from T_i to T_{i+1} . By measuring the phase-frequency responses at multiple steady-state temperature points, several $S_T(f)$ curves corresponding to adjacent temperature intervals can be obtained, enabling analysis of the dependence of the TCPS on both temperature and frequency.

2.2. TCPS Measurement Method

Based on the above definitions, a VNA is employed to perform swept-frequency phase measurements on the TIA under test. The VNA outputs a swept-frequency signal over a specified frequency range while simultaneously measuring the phase-frequency response of the device. By repeating these measurements at multiple steady-state temperature points, the phase-frequency responses $\varphi(f, T_1)$, $\varphi(f, T_2)$, $\varphi(f, T_3)$, $\varphi(f, T_4)$, and $\varphi(f, T_5)$ are obtained. The TCPS curves $S_{T,1}(f)$, $S_{T,2}(f)$, $S_{T,3}(f)$, and $S_{T,4}(f)$ are then calculated from the phase-frequency differences between adjacent temperature points.

Through this approach, the frequency response of the DUT's TCPS across the target frequency band can be derived. Compared with point-by-point measurements using a phasemeter, this method does not require signal configuration, phase locking, and phase readout to be performed separately at each frequency point. Instead, the phase data over the entire frequency band are obtained directly from the swept phase-frequency response, thereby improving the convenience and efficiency of frequency-domain TCPS measurements. Meanwhile, to avoid additional phase drift introduced during temperature transitions, the phase-frequency response at each temperature point is acquired only after thermal equilibrium has been reached.

An experimental test system based on a VNA and overall temperature modulation was established, as shown in Fig. 1.

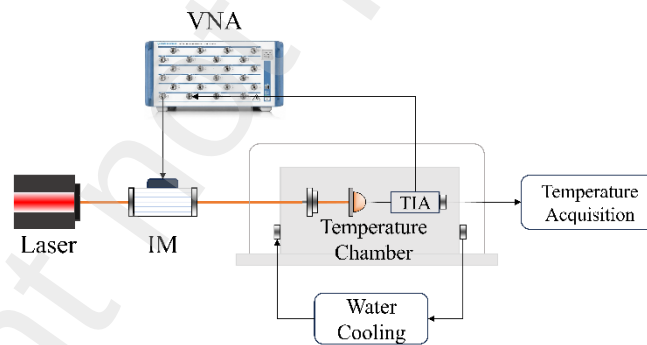


Fig. 1. Schematic diagram of the test system. (VNA: vector network analyzer; IM: intensity modulator; TIA: transimpedance amplifier)

The experimental setup consists of a laser, a fiber intensity modulator, a VNA, the TIA under test, an actively temperature-controlled chamber, and a temperature acquisition module. The actively controlled chamber serves to regulate the overall temperature of the TIA. A temperature probe inside the chamber provides closed-loop control, enabling the internal temperature to gradually stabilize at the vicinity of the set point. Meanwhile, an external temperature probe is positioned near the DUT to record the actual temperature at the circuit location. Given that a certain temperature difference may exist between the chamber control temperature and the temperature near the device, the steady-state temperature recorded by the external probe is used as the circuit

temperature when calculating the TCPS. By employing overall temperature modulation, parameters within the DUT, such as the feedback network and active components, vary with temperature, enabling direct measurement of the TCPS curve of the entire circuit.

3. Model of the DUT

To validate the proposed frequency-domain measurement method for the TCPS, a discrete-transistor-based TIA was selected as a representative photoreceiver front-end for experimental investigation. Traditional TIAs based on integrated operational amplifiers feature simple structures and mature applications. However, in MHz-level weak-light heterodyne detection scenarios, their bandwidth, noise, and stability are typically constrained by the operational amplifier's input noise, input capacitance, gain-bandwidth product, and feedback network parameters [17, 18]. In contrast, a discrete-transistor-based TIA offers greater design flexibility, allowing key design parameters such as the collector current and transconductance of the input transistor, as well as the feedback network parameters, to be adjusted according to specific requirements.

The transistor-based TIA circuit used in this work mainly follows the discrete heterojunction bipolar transistor (HBT) photoreceiver scheme proposed by Fernández Barranco et al. [15]. Designed for inter-satellite laser interferometry, this scheme uses HBTs to construct the transimpedance front-end and achieves low input-referred current noise and a large bandwidth in the MHz-level heterodyne frequency band. These results indicate the potential of transistor-based TIAs for high-precision laser interferometric optoelectronic readout links [15]. Meanwhile, the operating state of a transistor-based TIA depends on device parameters and bias conditions, and its phase-frequency response is highly sensitive to temperature variations. Therefore, this circuit was selected as the experimental device for this study. We measured its phase-frequency responses under different temperature conditions to obtain the frequency response of the TCPS and to further evaluate the feasibility and repeatability of the proposed frequency-domain measurement method.

In the experiment, the transistor-based TIA under test used a feedback resistance of $R_F=55\text{ k}\Omega$. The VNA swept the frequency from 1 MHz to 50 MHz with 1000 sampling points. Five steady-state temperature points were selected. For each temperature point, seven complex frequency-response traces were recorded and averaged in the complex domain before extracting the phase-frequency response. An external temperature probe positioned near the DUT recorded the temperature data at a sampling rate of 1 Hz. The steady-state temperature recorded by this probe was used as the circuit temperature in the calculation of the TCPS.

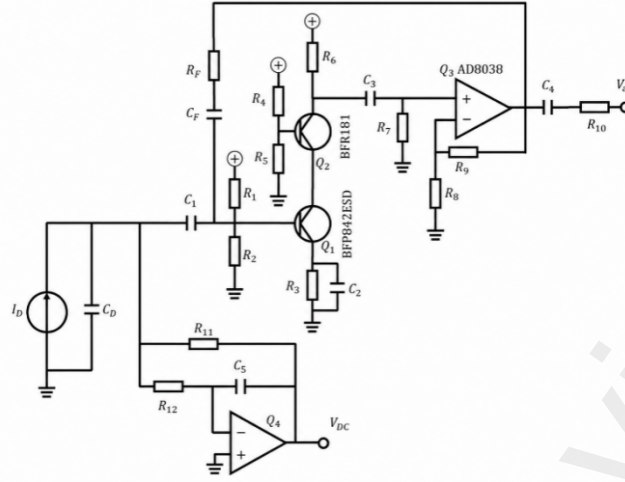


Fig. 2. Schematic of the transistor-based TIA.

To further analyze the validity of the experimental measurements, a total transfer-function model of the transistor-based TIA under test was established. In small-signal AC analysis, the photodiode is modeled as a parallel combination of a photocurrent source and junction capacitance C_D . Transistors Q_1 and Q_2 are described using a high-frequency hybrid- π model [19], in which the base-emitter equivalent resistance, transconductance, and junction capacitance collectively determine the frequency response of the common-emitter–common-base cascaded circuit. The feedback loop consists of a feedback resistor and a feedback capacitor, whose impedance directly affects the transimpedance gain and bandwidth of the circuit. Based on this equivalent model, the overall transfer function of the circuit's AC channel can be derived.

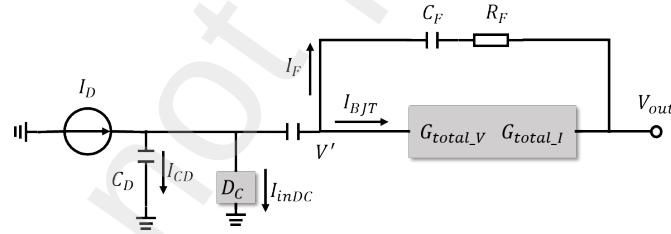


Fig. 3. Schematic diagram for calculating the overall transfer function of the transistor-based TIA.

When establishing the transfer function for the AC path, rather than expanding the complete nodal equations within the common-emitter common-base cascode circuit, this paper models the circuit together with the overall transfer function of the subsequent voltage amplifier as a voltage-input open-loop transfer function G_{total_V} and a current-input open-loop transfer function G_{total_I} . Here, G_{total_V} and G_{total_I} are obtained by multiplying the transfer function of the common-emitter common-base cascaded circuit by that of the post-stage voltage amplifier, respectively:

$$G_{total_V} = G_{cascode_V} G_{opamp\#} \quad (5)$$

$$G_{total_I} = G_{cascode_I} G_{opamp\#} \quad (6)$$

Based on the equivalent structure shown in Fig. 3, the AC photocurrent generated by the photodiode can be divided into currents flowing through the junction capacitance, the DC path, the feedback loop, and the transistor AC channel. Accordingly, we have:

$$I_D = I_{CD} + I_{inDC} + I_F + I_{BJT} \# (7)$$

Thus, the overall transfer function from the circuit input current to the output voltage can be obtained as:

$$G_{total} = \frac{1}{\frac{F}{Z_{CD}} + Y_{DC}F + \frac{1}{G_{total_I}} + \frac{\frac{1}{G_{total_V}} - 1}{Z_F}} \# (8)$$

where

$$F = Z_{c1} \left(\frac{1}{G_{total_I}} + \frac{\frac{1}{Z_{c1}} + \frac{1}{Z_F}}{G_{total_V}} - \frac{1}{Z_F} \right) \# (9)$$

$$Z_F = R_F + \frac{1}{sC_F} \# (10)$$

$$Z_{CD} = \frac{1}{sC_D} \# (11)$$

where Z_F is the impedance of the feedback loop, Z_{CD} is the impedance of the photodiode junction capacitance, Y_{DC} represents the equivalent input admittance of the DC path for AC signals, and $s = j2\pi f$.

4. Experimental Results

The TCPS curves are obtained from the difference between the phase-frequency responses measured at different steady-state temperature points. Therefore, the stability of the temperature points directly affects the calculation of the TCPS. During the experiment, an external temperature probe was placed near the DUT to record the actual temperature at the circuit location. Fig. 4. shows the temperature variation recorded by the external temperature probe during the experiment, where the gray shaded regions indicate the time windows used for steady-state temperature statistics and phase-frequency response acquisition.

Since the time required to reach thermal equilibrium differs among set temperature points, the stabilization time for each point varies. Based on the temperature data recorded by the external temperature probe, steady-state statistical windows with small temperature variations were selected within each temperature plateau. The mean temperature within each window was used as the actual steady-state temperature of the corresponding temperature point, and the standard deviation was used to represent the temperature fluctuation at that point. In the subsequent calculation of the TCPS, these actual steady-state temperature values were used rather than the chamber set temperatures or the internal closed-loop control temperatures.

The five temperature points were 27.456(8) °C, 29.103(2) °C, 30.784(7) °C, 32.541(6) °C, and 34.318(4) °C. The temperature fluctuations within each steady-state statistical window were small, indicating good stability of the temperature near the DUT during the phase-frequency response acquisition. These measurements can therefore be used for subsequent calculation of the TCPS.

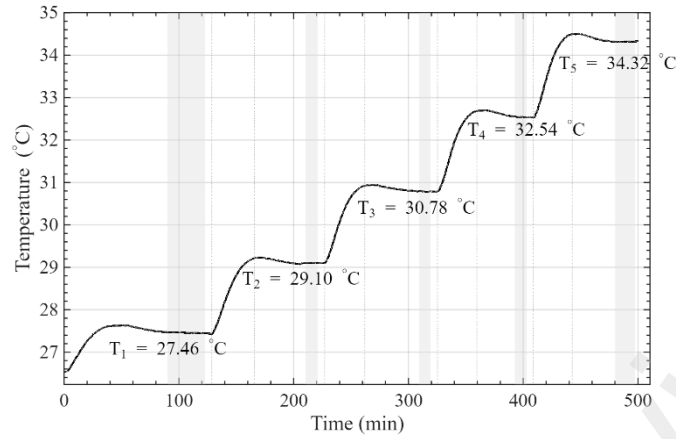


Fig. 4. Temperature variation recorded by the external temperature probe.

During temperature adjustment, VNA measurement data were recorded once the chamber temperature fluctuations met the preset stability criteria. Fig. 5. shows the measured phase-frequency response curves at the five temperature points. The main plot presents the overall phase-frequency response in the 1-50 MHz range, while the inset plot provides a locally magnified view around 20 MHz.

As shown in Fig. 5, the overall trend of the phase-frequency response curves at different steady-state temperatures is similar, with slight differences appearing in local frequency bands. Since the phase drift caused by temperature variation is small relative to the total phase change, differences among curves corresponding to different temperature points are not apparent in the full-frequency phase-frequency response plot. Local magnification reveals resolvable phase differences in specific frequency bands, and these differences vary with frequency.

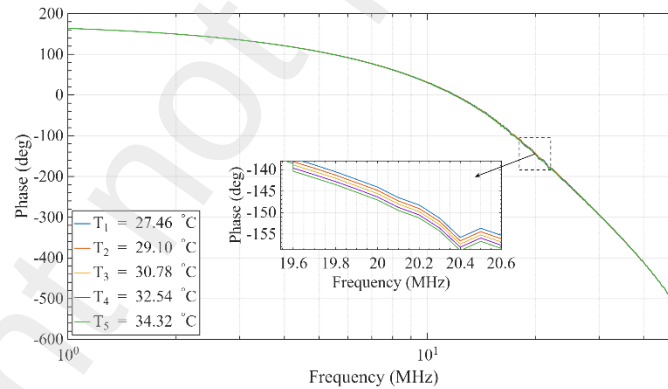


Fig. 5. Experimental phase-frequency response curves of the TIA under test at different steady-state temperatures.

The above results indicate that temperature variations induce changes in the phase-frequency response of the DUT. However, these changes are not simple frequency-independent offsets; rather, they are frequency-dependent. Therefore, measuring the phase variation with temperature at a single frequency point cannot fully capture the thermally induced phase response of the circuit across the entire target frequency band. To accurately characterize the temperature-dependent phase behavior of the device, frequency-domain measurements and analysis over the target band are necessary.

Based on the phase-frequency responses measured at different steady-state temperature points, the TCPS of the DUT was further calculated using the phase difference between adjacent temperature points. Fig. 6. shows the resulting TCPS curves in the 1-50 MHz range. The four curves correspond to the adjacent temperature intervals T_2-T_1 , T_3-T_2 , T_4-T_3 , and T_5-T_4 , respectively. Each curve is derived by calculating the difference of phase-frequency responses between adjacent steady-state temperature points, which reflects the sensitivity of the circuit phase to temperature variations within each corresponding interval.

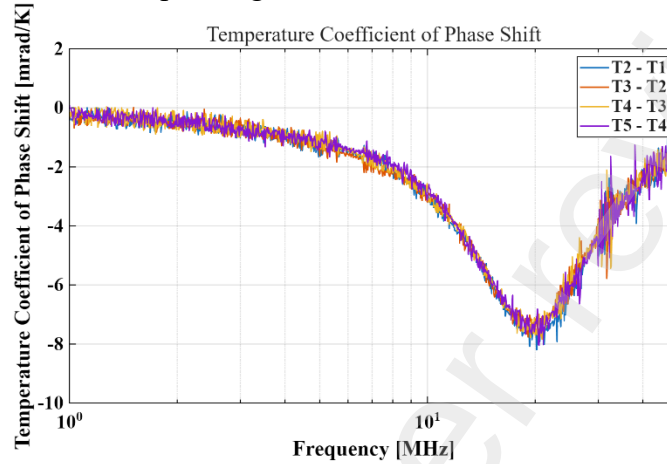


Fig. 6. Measured TCPS curves.

It can be seen that the TCPS of the transistor-based TIA under test is not constant over the 1-50 MHz range, but varies significantly with frequency. Statistical analysis of repeated measurements for each temperature interval shows that the extremum frequencies for all intervals are concentrated near 20 MHz, with extremum values ranging from approximately -7.4 to -7.6 mrad/K. Considering all temperature intervals and repeated measurements, the extremum frequency is 19.8 ± 0.2 MHz, corresponding to an extremum of -7.5 ± 0.2 mrad/K. At low frequencies, the absolute value of the TCPS is small; it gradually increases with frequency, reaching a maximum near 20 MHz, and then decreases at higher frequencies. This indicates that the thermally induced phase response of the device exhibits a clear frequency-dependent behavior, with phase sensitivity to temperature perturbations primarily concentrated near specific frequency regions.

Table 1. Measurement results of the TCPS.

Temperature interval	Extremum frequency / MHz	Extremum value / mrad/K
T_2-T_1	20.0 ± 0.3	-7.6 ± 0.2
T_3-T_2	19.7 ± 0.2	-7.5 ± 0.2
T_4-T_3	19.9 ± 0.2	-7.5 ± 0.1
T_5-T_4	19.7 ± 0.3	-7.4 ± 0.2

5. Discussion

The TCPS calculated based on nominal parameters can be used to analyze the frequency-response characteristics of the DUT. However, actual circuit components, such as resistors, capacitors, and small-signal transistor parameters, may deviate from their nominal values. These parameter uncertainties further affect the phase-frequency response and the TCPS. Therefore, a point-by-point comparison between the nominal model and experimental results is insufficient. To evaluate the influence of parameter uncertainties on the frequency response of the TCPS, a Monte Carlo simulation was conducted based on the complete circuit model [19, 20].

In the Monte Carlo simulation, random perturbations were applied to the model parameters affecting the phase-frequency response, centered on their nominal values. The perturbation ranges were set as $\pm 10\%$ for the small-signal transistor parameters, $\pm 5\%$ for the bias currents, $\pm 10\%$ for the post-stage amplifier parameters, and $\pm 10\%$ for the temperature coefficients. All parameters were sampled randomly within their respective predefined ranges according to a uniform distribution [21].

For each set of randomized parameters, the phase-frequency responses at different temperature points were recalculated, and the corresponding TCPS curves were obtained using the phase-difference method between adjacent temperature points. After 1000 random simulations, the statistical distribution of the TCPS at different frequencies was obtained. To reduce the influence of extreme random samples, the 5% - 95% interval was adopted to represent the distribution range of the Monte Carlo simulation results.

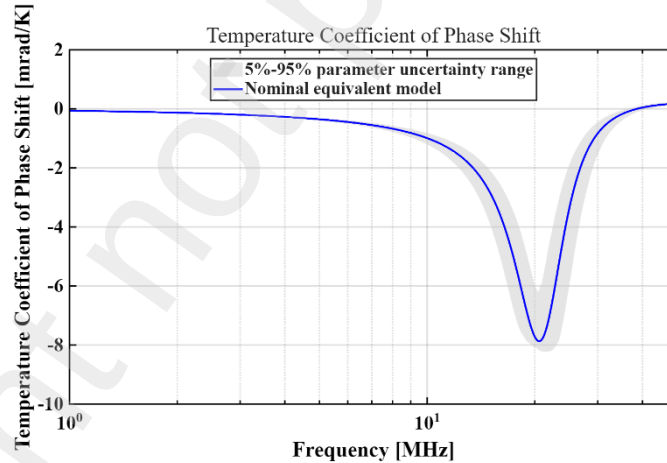


Fig. 7. Monte Carlo simulation results of the TCPS curves under parameter uncertainties.

The experimentally measured TCPS curve is further compared with the Monte Carlo simulation results. As shown by the results, the experimental curve and the simulated frequency-response distribution exhibit similar overall trends, both showing a pronounced extremum near 20 MHz. Considering the effects of parasitic parameters in the actual circuit, deviations of device parameters from their nominal values, non-uniform temperature distribution, and the difference between the temperature recorded by the external temperature probe and the actual junction temperature of the devices, the circuit-level model cannot be expected to reproduce the experimental curve point by point over the entire frequency range. Therefore, this work mainly selects the

extremum frequency and extremum magnitude as two characteristic parameters for comparing the experimental and simulation results.

The extremum frequency of the TCPS is mainly affected by the circuit pole-zero locations and their temperature sensitivities, while the extremum magnitude reflects the sensitivity of the phase response to temperature variations near the characteristic frequency. The experimentally extracted extremum frequency is 19.8 ± 0.2 MHz, which falls within the simulated extremum-frequency distribution range of 19.0–22.2 MHz. The experimentally obtained extremum magnitude is -7.5 ± 0.2 mrad/K, which also falls within the simulated extremum-magnitude distribution range of -8.1 to -7.3 mrad/K. These results indicate that, after considering the uncertainties of device parameters and parasitic parameters, the simulation can provide an extremum region and magnitude comparable to the experimental results. Together with the overall trend of the frequency-response curves, the comparison suggests that the established model can explain the main frequency-dependent features of the temperature-to-phase-shift coefficient of the transistor-based TIA, and that the proposed frequency-domain measurement method can effectively extract key characteristic parameters of the thermally induced phase response.

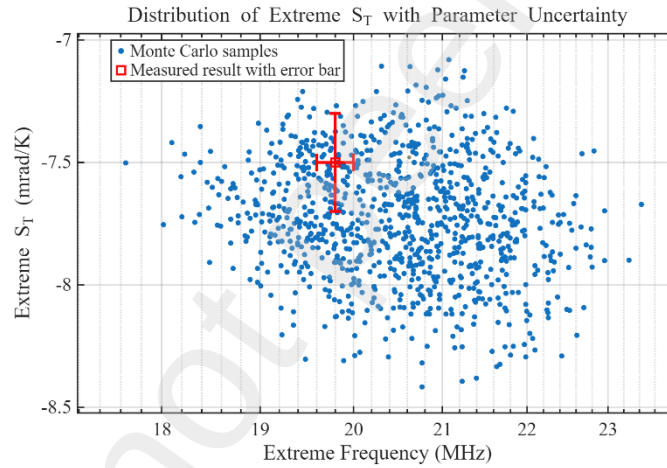


Fig. 8. Distributions of the extremum TCPS value and frequency from Monte Carlo simulations, with the measured result indicated by the hollow square.

Based on the combined experimental and simulation results, the proposed measurement method, which utilizes swept phase-frequency responses and overall temperature modulation, can obtain the frequency response of the TCPS of the TIA under test over the target frequency band. Compared with single-frequency-point measurements, this method not only provides the temperature-induced phase sensitivity at specific frequencies, but also enables the identification of the distribution characteristics of the TCPS with frequency, sensitive frequency bands, and extremum locations. For MHz-level heterodyne optoelectronic readout links, such frequency-domain information facilitates a more comprehensive assessment of thermally induced phase noise and provides guidance for subsequent circuit parameter selection and thermal stability design.

6. Conclusion

In this study, a frequency-domain measurement method for the TCPS based on VNA-swept phase-frequency response and overall temperature modulation is proposed. In this method, the phase-frequency responses of the photoreceiver front-end under test are measured at different steady-state temperature points, and the TCPS curves over the target frequency range are obtained by phase differencing between adjacent temperature points. Compared with conventional single-frequency measurement methods, the proposed method can directly obtain the frequency-continuous characterization of the TCPS, thereby providing more complete experimental information for evaluating the thermal stability of broadband optoelectronic readout chains.

A discrete-transistor-based TIA was used as a representative photoreceiver front-end, and frequency-domain measurements of the TCPS were carried out over 1-50 MHz. The experimental results show that the TCPS of the circuit under test exhibits clear frequency dependence and reaches an extremum near 20 MHz. The measured extremum frequency is 19.8 ± 0.2 MHz, and the corresponding extremum magnitude is -7.4 ± 0.2 mrad/K, indicating that the proposed method can effectively extract the sensitive frequency region and key characteristic parameters of the thermally induced phase response of the circuit.

Furthermore, an AC transimpedance transfer-function model of the circuit under test was established, and Monte Carlo simulations were performed to analyze the influence of parameter uncertainties on the temperature-to-phase-shift coefficient. The experimental results show good agreement with the Monte Carlo simulation results in terms of the key features of extremum frequency and extremum magnitude, indicating that the established model can provide a reasonable explanation for the experimentally observed frequency-dependent behavior and extremum region. The proposed method can provide a reference for thermally induced phase-drift evaluation, phase-noise budget analysis, and thermal-stability design of photoreceiver front-ends in high-precision laser interferometry, optoelectronic readout chains, and broadband heterodyne detection systems.

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